

Cuanify Complete Documentation

Dokumen ini adalah referensi teknis dan operasional lengkap untuk aplikasi Cuanify (frontend, backend API, database, integrasi eksternal, deployment, dan runbook).

Dokumen pendamping:

- User Journey: `SOP/Cuanify-User-Journey.md`

1. Ringkasan Produk

Cuanify adalah platform POS + inventory + analytics berbasis AI untuk UMKM (terutama F&B/bakery/retail).

Aplikasi menyediakan:

- Pencatatan transaksi dan kasir shift.
- Manajemen produk, kategori, bahan baku, resep, dan stok.
- Analitik bisnis, forecasting, dan insight AI.
- Flow bakery booking end-to-end (order, shipping quote/resi, automasi WhatsApp/Google Calendar/Sheets).
- Integrasi pembayaran (Midtrans dan Xendit).
- Asisten AI dengan RAG berbasis dokumen dan data bisnis internal.

2. Arsitektur Sistem

2.1 Arsitektur High-Level

- Frontend dan backend berada dalam satu aplikasi Next.js (App Router).
- API routes berada pada `client-side/app/api/**`.
- Data utama disimpan di PostgreSQL via Prisma.
- Embedding RAG menggunakan pgvector (`vector` extension).
- Integrasi eksternal dipanggil dari server-side route/service.

2.2 Komponen Utama

- UI/SSR/Server Actions: Next.js 16 + React 19.
- API Layer: Next.js Route Handler.
- ORM/DB: Prisma + `@prisma/adaptor-pg` + pool PG.
- Auth: NextAuth JWT + custom JWT fallback.
- AI: Groq (dan dukungan env Gemini).
- Integrasi: Fonnte, Google Calendar, Google Sheets, Biteship, Midtrans, Xendit, ImageKit.

3. Tech Stack

3.1 Frontend

- Next.js 16
- React 19
- TypeScript 5
- Tailwind CSS 4
- react-hook-form + zod
- recharts, framer-motion, sonner, lucide-react

3.2 Backend

- Next.js API Routes

- Prisma 7 + `@prisma/client`
- PostgreSQL (`pg`)
- next-auth, jsonwebtoken, bcryptjs
- nodemailer
- groq-sdk

3.3 Data dan AI

- PostgreSQL + pgvector
- pdf-parse untuk ekstraksi dokumen

4. Struktur Proyek

4.1 Workspace

- `README.md` : profil produk global.
- `client-side/` : aplikasi utama Next.js.
- `data/` : data seed/reference JSON.
- `SOP/` : dokumen operasional/go-live.

4.2 Struktur App Utama (`client-side`)

- `app/` : page/layout/error + API routes.
- `components/` : komponen UI domain.
- `context/` : context provider aplikasi.
- `lib/` : domain service/helper/integrasi.
- `prisma/` : schema + migrations.
- `scripts/` : utilitas setup, readiness, seed, smoke test.
- `public/` : static assets + PWA manifest/SW.

5. Domain Fitur dan Modul

5.1 Authentication & Authorization

- Endpoint auth ada di `app/api/auth/**` .
- Resolusi user menggunakan:
 - NextAuth JWT (`next-auth/jwt`).
 - Fallback custom JWT (`token` `cookie` / Bearer).
- Resolusi business aktif:
 - Prioritas cookie `active_business_id` .
 - Fallback ke business owner pertama.
 - Fallback membership kasir (`BusinessMember`).

5.2 POS, Inventory, Recipe, Production

- Produk, kategori, bahan baku, dan recipe terhubung dengan transaksi sales.
- Movement stok mengikuti flow restock/consume/sale.
- Produksi dan batch tersedia untuk kebutuhan manufaktur ringan bakery/F&B.

5.3 Analytics dan Forecasting

- API analytics berada pada `app/api/analytics/**` .
- Tersedia dashboard metrik, growth, health, hourly/daily/monthly, top products, waste, forecasting.
- Ada endpoint cached untuk optimasi response tertentu.

5.4 Debts (Kasbon/Piutang)

- Manajemen hutang pelanggan dan pembayaran cicilan.
- Endpoint ada di `app/api/debts/**` .

5.5 Bakery Booking & Automations

- Domain file utama: `lib/bookings/**` , `components/bakery/**` , `app/api/bookings/**` .
- Fitur utama:
 - Parse order dari chat/image WhatsApp.
 - Kalkulasi harga, add-on, DP, slot delivery.
 - Rule pengiriman (termasuk item tertentu GrabCar only).
 - Quote shipping + pembuatan resi via Biteship.
 - Persist order via API `bookings/orders` .
 - Automasi notifikasi WhatsApp produksi/customer.
 - Integrasi Google Calendar dan Google Sheets.

5.6 AI Assistant (RAG)

- Endpoint AI ada di `app/api/ai/**` .
- Subdomain:
 - Chat sessions.
 - RAG index/upload/search/documents.
 - Insight generation.

6. Database Model (Ringkas)

Skema utama ada di `client-side/prisma/schema.prisma` .

6.1 Entitas Core

- `User`
- `Business`
- `BusinessMember` (RBAC Owner/Cashier)

6.2 Entitas Operasional

- `Product` , `Category` , `Ingredient`
- `Sale` dan detail turunan
- `StockDocument` , inventory movements
- `CashierShift`
- `Debt` , `DebtPayment`

6.3 Entitas AI/RAG

- `BusinessDocument` (content, metadata, embedding vector)
- `ChatSession` , `ChatMessage`

6.4 Catatan Khusus Bakery Orders

- Persist order bakery menggunakan tabel khusus row-based (`bakery_orders` , `bakery_order_items` , `bakery_order_addresses`) plus snapshot fallback melalui `BusinessDocument` .
- Pendekatan ini dipakai untuk sinkronisasi cepat UI booking dan resilience fallback.

7. API Catalog

7.1 AI

- /api/ai/chat
- /api/ai/insights
- /api/ai/sessions
- /api/ai/rag/documents
- /api/ai/rag/index
- /api/ai/rag/search
- /api/ai/rag/upload

7.2 Analytics

- /api/analytics/category
- /api/analytics/daily
- /api/analytics/dashboard
- /api/analytics/debts-stats
- /api/analytics/forecast
- /api/analytics/forecast-cached
- /api/analytics/growth
- /api/analytics/health
- /api/analytics/hourly
- /api/analytics/insight
- /api/analytics/insights-cached
- /api/analytics/last-updated
- /api/analytics/monthly
- /api/analytics/products
- /api/analytics/top-products
- /api/analytics/waste

7.3 Auth

- /api/auth/[...nextauth]
- /api/auth/login
- /api/auth/logout
- /api/auth/me
- /api/auth/post-login
- /api/auth/profile
- /api/auth/profile/password
- /api/auth/register

7.4 Bookings

- /api/bookings/automations
- /api/bookings/catalog-config
- /api/bookings/orders
- /api/bookings/parse-whatsapp
- /api/bookings/marketplace-email/latest
- /api/bookings/shipping/quote
- /api/bookings/shipping/create-resi
- /api/bookings/google-calendar/connect

- /api/bookings/google-calendar/callback
- /api/bookings/google-calendar/status
- /api/bookings/google-calendar/events
- /api/bookings/google-calendar/disconnect

7.5 Business dan Master Data

- /api/businesses
- /api/businesses/[id]
- /api/business-analytics
- /api/categories
- /api/products
- /api/products/[id]
- /api/products/[id]/recipe
- /api/products/generate/name
- /api/products/generate/image
- /api/products/generate/recipe-image
- /api/products/generate/recommend-price
- /api/ingredients
- /api/ingredients/[id]
- /api/ingredients/[id]/restock
- /api/ingredients/[id]/consume

7.6 Sales, Payment, Staff, Operations

- /api/sales
- /api/sales/[saleId]/confirm
- /api/sales/[saleId]/invoice
- /api/sales/midtrans-token
- /api/sales/midtrans-notification
- /api/debts
- /api/debts/[id]/pay
- /api/cashier-shift
- /api/cashier-shift/open
- /api/cashier-shift/close
- /api/cashier-shift/current
- /api/staff
- /api/staff/register
- /api/production

7.7 Support / Utility

- /api/export/inventory
- /api/export/sales
- /api/analyze-image
- /api/cleanup-images
- /api/cron/generate-analytics
- /api/mock-business
- /api/mock-business/create
- /api/mock-ingredients
- /api/debug/business/[id]

8. Integrasi Eksternal

8.1 WhatsApp (Fonnte)

- Digunakan untuk notifikasi produksi/customer pada flow booking.
- Konfigurasi env:
 - FONNTE_TOKEN
 - FONNTE_PRODUCTION_TARGET
 - FONNTE_SEND_PRODUCTION_ON_CREATE
 - FONNTE_SEND_CUSTOMER_ON_CONFIRM
 - FONNTE_NOTIFY_RESCHEDULE_PRODUCTION
- Catatan penting:
 - Tetap membutuhkan sender nomor WhatsApp aktif.
 - FONNTE_PRODUCTION_TARGET dapat nomor atau group id `...@g.us`.

8.2 Google Calendar / Sheets

- Calendar dipakai untuk event jadwal booking.
- Sheets dipakai sinkronisasi order operasional.
- Mendukung OAuth callback + service account fallback sesuai konfigurasi.

8.3 Shipping (Biteship)

- Quote ongkir dan create resi.
- Provider aktif saat ini mencakup JNE, J&T, Paxe (tergantung konfigurasi/coverage).

8.4 Payment

- Midtrans untuk tokenisasi dan notification payment gateway.
- Xendit untuk flow invoice tambahan.

8.5 Media dan AI

- ImageKit untuk asset gambar.
- Groq API untuk insight AI.

9. Environment Variables

Sumber acuan utama: `client-side/.env.example`.

9.1 Wajib Minimum (Baseline App)

- NODE_ENV
- NEXTAUTH_URL
- NEXTAUTH_SECRET
- JWT_SECRET
- DATABASE_URL

9.2 Booking Automation

- FONNTE_TOKEN
- FONNTE_PRODUCTION_TARGET
- GOOGLE_CALENDAR_OAUTH_REDIRECT_URI
- GOOGLE_CALENDAR_ID
- GOOGLE_SERVICE_ACCOUNT_EMAIL

- `GOOGLE_SERVICE_ACCOUNT_PRIVATE_KEY`
- `GOOGLE_SHEETS_ID`
- `GOOGLE_SHEETS_RANGE`

9.3 Shipping

- `BITESHIP_API_KEY`
- `SHIPPING_ORIGIN_ADDRESS`
- `SHIPPING_ORIGIN_POSTAL_CODE`
- `SHIPPING_ORIGIN_LATITUDE`
- `SHIPPING_ORIGIN_LONGITUDE`
- `SHIPPING_ORIGIN_CONTACT_NAME`
- `SHIPPING_ORIGIN_CONTACT_PHONE`
- `SHIPPING_ORIGIN_CONTACT_EMAIL`

9.4 Payment

- Midtrans:
 - `MIDTRANS_SERVER_KEY`
 - `MIDTRANS_CLIENT_KEY`
 - `NEXT_PUBLIC_MIDTRANS_CLIENT_KEY`
 - `MIDTRANS_IS_PRODUCTION`
 - `NEXT_PUBLIC_MIDTRANS_IS_PRODUCTION`
- Xendit:
 - `XENDIT_SECRET_KEY`

9.5 AI, Email, Storage, Cron

- `GROQ_API_KEY` , `GEMINI_API_KEY`
- `EMAIL_HOST` , `EMAIL_PORT` , `EMAIL_SECURE` , `EMAIL_USER` , `EMAIL_PASSWORD` , `EMAIL_FROM`
- `IMAGEKIT_PUBLIC_KEY` , `IMAGEKIT_PRIVATE_KEY` , `IMAGEKIT_URL_ENDPOINT`
- `CRON_SECRET` , `ENABLE_AUTO_CLEANUP` , `LOG_LEVEL`

9.6 Public Bakery Config

- `NEXT_PUBLIC_BAKERY_DOWN_PAYMENT_PERCENT`
- `NEXT_PUBLIC_BAKERY_BLOCKED_DATES`

10. Setup Development

Dari folder `client-side` :

1. Install dependency:
 - `npm install`
2. Siapkan env:
 - copy `.env.example` ke `.env`
 - isi variabel minimum
3. Prisma generate + migrate sesuai kebutuhan DB.
4. Jalankan dev server:
 - `npm run dev`

11. Build, Deploy, dan Readiness

11.1 Build

- `npm run build`

11.2 Start Production Mode Lokal

- `npm run start`

11.3 Readiness Check

- Sandbox: `npm run check:ready:sandbox`
- Production: `npm run check:ready:prod`

Readiness checker memvalidasi env wajib, mode Midtrans, dan warning operasional (mis. blocked dates dan Google Sheets).

11.4 Deploy Vercel

- Build command di `vercel.json` :
 - `npx prisma generate && npx next build --webpack`
- Cron terdaftar:
 - `/api/cron/generate-analytics` pada `0 3 * * *`

12. Runbook Operasional

12.1 Daily Checks

- Pastikan login/auth normal.
- Cek transaksi POS masuk ke dashboard analytics.
- Cek stok bahan baku berkurang sesuai sales.
- Untuk bakery:
 - Cek blocked dates, slot delivery, quote shipping.
 - Cek automasi WA/Calendar/Sheets berjalan.

12.2 Incident Response (Umum)

1. Cek Vercel logs (status code, stack trace).
2. Cek env di Vercel Project Settings.
3. Validasi koneksi DB dan schema/migration.
4. Jalankan smoke script terkait (`test-email` , `xendit:smoke` , `dst`).
5. Jika perlu, rollback ke deployment stable terakhir.

12.3 Kasus Bakery Orders Gagal Persist

Gejala:

- `POST /api/bookings/orders` status 500.

Kemungkinan penyebab:

- Transaction timeout terlalu kecil untuk payload besar.
- Koneksi DB lambat atau lock contention.
- Env DB tidak stabil.

Mitigasi yang sudah diterapkan:

- Timeout interactive transaction diperbesar pada endpoint `orders sync`.

13. Security & Compliance Checklist

- Jangan commit `.env` ke repository publik.
- Rotasi secret jika pernah terekspos (WA token, DB URL, OAuth secret, payment keys).
- Gunakan channel WA resmi/berizin dan opt-in pelanggan.
- Batasi akses endpoint sensitif dengan auth/role check.
- Pastikan webhook payment tervalidasi signature.
- Gunakan principle of least privilege untuk service account Google.

14. Script Operasional Penting

- `npm run check:ready:sandbox`
- `npm run check:ready:prod`
- `npm run seed:mock-sales`
- `npm run test:email`
- `npm run xendit:smoke`
- Script utilitas lain ada di folder `client-side/scripts`.

15. Known Constraints

- Integrasi WhatsApp outbound tetap membutuhkan sender nomor WA bisnis aktif.
- Beberapa fitur automation bergantung pada konfigurasi env lengkap; jika env kosong, fitur akan skip/fallback.
- Performa endpoint sinkronisasi data besar sensitif terhadap latency DB.

16. Rekomendasi Lanjutan

1. Tambahkan dokumentasi OpenAPI/typed contract untuk endpoint prioritas.
2. Tambahkan test integrasi untuk alur booking end-to-end (orders + automations + shipping).
3. Tambahkan observability terstruktur (request id, business id, latency tracing).
4. Buat runbook recovery khusus per integrasi (Midtrans, Fonnte, Biteship, Google API).

Dokumen ini harus di-update setiap ada perubahan arsitektur, endpoint kritikal, env, atau flow operasional.